

WASP-12b

R-0341

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231208 · created 2026-07-18T07:01:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460286.9516 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.044
Transit depth (Rp/Rs)^2	7.5e-06 +/- 0.0024 [%]
Orbital Inclination (inc)	86.55 +/- 2.91
Airmass coefficient 1 (a1)	1.14 +/- 0.26
Airmass coefficient 2 (a2)	-0.02 +/- 0.11
Scatter in the residuals of the lightcurve fit is	22.58 %
Best Comparison Star	None
Optimal Aperture	3.38
Optimal Annulus	5.28
Transit Duration (day)	0.1173 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.044 vs 0.1178 (deviation 2.68σ)
Duration fit vs published	0.1173 d vs 0.125 d (deviation 1.54σ)
Mid-time O–C	0.4 min
Depth SNR	0.0
Residual scatter	22.58 %

Light curve

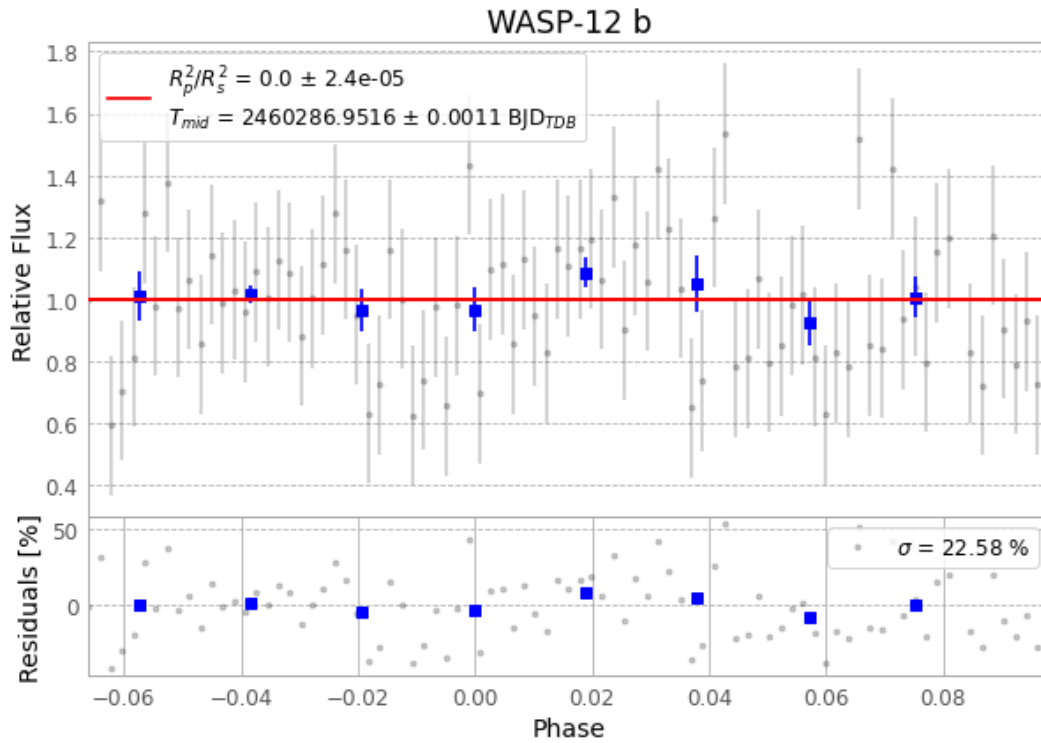


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-08.png (SHA-256 48ac96bd60f4b440...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 256], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 62373419670fd119...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

87 FITS frames (57 MB), WASP-12231208085715.FITS → WASP-12231208131515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231208.log (e3d6c3e47a79...), FinalLightCurve_WASP-12 b_2023-12-08.png (48ac96bd60f4...), FinalLightCurve_WASP-12 b_2023-12-08.csv (c35d5d9740be...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 07:01Z.